Andre He

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Ph.D. student, Carnegie Mellon University

andrehe02.github.io

Research interests Language-model reasoning, reinforcement learning, and efficient inference.

Recent work studies latent-token reasoning in diffusion models, synthetic environments with verifiable rewards, and reinforcement learning beyond distribution sharpening.

Education

Carnegie Mellon University — Fall 2024 – present
PhD in Language Technologies Institute (LTI)
Co-advisors: Sean Welleck, Daniel Fried

University of California, Berkeley — Aug 2020 – May 2024
B.A. Computer Science
Major GPA: 4.0

Work

Amazon AWS AI, New York, NY — May 2025 - Aug 2025
Applied Scientist Intern
Synthetic Data Generation for LLM Reasoning
Mentors: Nathaniel Weir, Kaj Bostrom

Robotic AI & Learning Lab — Aug 2022 - Jan 2024
Undergraduate Researcher
Language-Conditioned Robotics, Foundation Model for Robotics
Advisor: Sergey Levine

Scale AI — May 2022 - Aug 2022
Software Engineer Intern

Berkeley NLP Group — Aug 2021 - 2023
Undergraduate Researcher
Computational Linguistics, Interpretability
Advisor: Dan Klein

Teaching

CS170: Efficient Algorithms and Intractable Problems — 2022
Teaching assistant

Talks

Rewarding the Unlikely: Lifting GRPO Beyond Distribution Sharpening — Nov 2025
Oral Presentation, EMNLP 2025

Advanced Modeling: Diffusion — Nov 2025
Guest Lecture, 11-711 Advanced Natural Language Processing (CMU)

Reasoning with Latent Tokens in Diffusion Language Models — Mar 2026
Discrete Diffusion Reading Group

Discrete Diffusion Models — Mar–Apr 2026
Flow Reading Group

Publications and Preprints

* denotes equal contribution.

Reasoning with Undecoded Tokens in Diffusion Language Models
Andre He, Sean Welleck, Daniel Fried
NeurIPS 2026 (Spotlight). [Paper]

ReSyn: Autonomously Scaling Synthetic Environments for Reasoning Models
Andre He, Nathaniel Weir, Kaj Bostrom, Allen Nie, Darion Cassel, Sam Bayless, Huzefa Rangwala
Preprint, 2026. [Paper]

AdaExplore: Failure-Driven Adaptation and Diversity-Preserving Search for Efficient Kernel Generation
Weihua Du, Jingming Zhuo, Yixin Dong, **Andre Wang He**, Weiwei Sun, Zeyu Zheng, Manupa Karunaratne, Ivan Fox, Tim Dettmers, Tianqi Chen, Yiming Yang, Sean Welleck
Preprint, 2026. [Paper]

SimpLM: Simplex-Based Flow Language Modeling via Autoregressive Adaptation
Ningyuan Yang, Weihua Du, Weiwei Sun, **Andre He**, Sean Welleck, Yiming Yang
NonAR-LM Workshop @ COLM 2026. [Paper]

Efficient Test-Time Adaptation through Human-AI Interaction
Zora Zhiruo Wang, Apurva Gandhi, Rulin Shao, Aspen Chen, Jonas Mueller, Zhiqi Liang, Jett Chen, Michael Ryan, Qianou Ma, Luxi He, Zhoujun Cheng, **Andre He**, Seungone Kim, Jiayi Geng, Mingqian Zheng, Weiwei Sun, Zheyuan Zhang, Xinran Zhao, Yike Wang, Abe Hou, Liwei Jiang, Pang Wei Koh, Diyi Yang, Graham Neubig, Daniel Fried
Preprint, 2026. [Paper]

Rewarding the Unlikely: Lifting GRPO Beyond Distribution Sharpening
Andre He, Daniel Fried, Sean Welleck
EMNLP 2025 (Oral). [Paper]

BridgeData V2: A Dataset for Robot Learning at Scale
Homer Walke, Kevin Black, Abraham Lee, Moo Jin Kim, Max Du, Chongyi Zheng, Tony Zhao, Philippe Hansen-Estruch, Quan Vuong, **Andre He**, Vivek Myers, Kuan Fang, Chelsea Finn, Sergey Levine
CoRL 2023. [Paper]

Goal Representations for Instruction Following: A Semi-Supervised Language Interface to Control
Vivek Myers*, **Andre He***, Kuan Fang, Homer Walke, Philippe Hansen-Estruch, Ching-An Cheng, Mihai Jalobeanu, Andrey Kolobov, Anca Dragan, Sergey Levine
CoRL 2023. [Paper]

Neural Unsupervised Reconstruction of Protolanguage Word Forms
Andre He, Nicholas Tomlin, Dan Klein
ACL 2023. [Paper]

Understanding Game-Playing Agents with Natural Language Annotations
Nicholas Tomlin, **Andre He**, Dan Klein
ACL 2022. [Paper]